

Alaa Aldeen

Founder & CEO, Nilus Lab | AI Engineer & Generative Biology Architect

Swansea & London, UK | San Diego, CA, USA

Phone: +44 7405 955116 | Email: alaa.aldeen@niluslab.com | Website: niluslab.com |

LinkedIn: [linkedin.com/in/alaa199439](https://www.linkedin.com/in/alaa199439)

PROFILE

AI engineer and biotechnology founder developing generative biology systems for cardiac research and therapeutic discovery. Architect of the Zenith single-cell platform, built around 2.42 million human cardiac cells and a 5,009-dimensional latent representation, with integrated computational biology and laboratory automation capabilities. Inventor and patent holder with international recognition, including a Silver Medal at the International Exhibition of Inventions Geneva. Background spanning AI engineering, clinical research, medical diagnostics and biotechnology innovation.

2.42M cardiac cells | 5,009D latent representation | Geneva Silver Medal | Patent holder

AI & MACHINE LEARNING ENGINEERING CAPABILITIES

- **Generative Foundation Models (scVI & PyTorch):** Architected PyTorch-based variational autoencoders (scVI) mapping 2.42M human single-cell transcriptomes into a 5,009D latent space manifold for cell-state trajectory inference.
- **Large Language Models & Bio-Transformers:** Integrated bio-LLM and transformer architectures for zero-shot gene perturbation prediction, mRNA sequence generation, and algorithmic codon optimization.
- **Structural AI & Protein Folding:** Incorporated structural AI models (AlphaFold 3, Boltz-1, ESMFold2, RFdiffusion) to simulate 3D protein folding, binding affinities, and factor interaction dynamics.
- **Custom Neural Loss Functions & Safety Controls:** Engineered custom PyTorch neural loss functions embedding biophysical safety constraints (OCT4 $\leq$ 0.35, c-MYC $\leq$ 0.30, >94% TNNT2 and Connexin-43 retention).
- **AI Data Annotation, Labelling & QA (2+ Yrs Exp):** 2+ years of hands-on experience in ML dataset annotation, semantic text/image tagging, RLHF quality assurance, and dataset curation for AI training models.
- **AI-Driven Robotics Automation (Opentrons OT-2):** Developed automated pipeline engines translating generative AI predictions into 1-click executable Python/CSV protocols for Opentrons OT-2 liquid-handling robots.
- **Multi-Agent AI Systems & Workflow Engines:** Architected Promatly.com, an enterprise multi-agent task orchestration platform running parallel LLM API jobs, webhooks, and RESTful microservices.
- **Data Pipeline Engineering & Biostatistics:** End-to-end scRNA-seq preprocessing data pipelines using Python (AnnData, Scanpy, NumPy, Pandas, SciPy, Scikit-Learn) and Biostatistics (SPSS, FlowJo).

WORK EXPERIENCE

NILUS LAB — London, United Kingdom | San Diego, California

Founder & Lead AI Engineer / CEO (2024 – Present)

- **Deep Neural Network Model Training:** Trained Zenith v1.0 GOLD, a 5,009D latent space single-cell variational autoencoder (scVI/PyTorch) across 2.42 million human cardiac single-cell transcriptomes.
- **AI-Guided Therapeutic Discovery (NL-101):** Applied generative AI models to discover NL-101, a non-oncogenic mRNA-LNP payload (SIRT1 + SIRT6 + GATA4 + ZBTB16) achieving a -13.0-year biological age reset measured via the Meyer-Schumacher BiT age (Binarized Transcriptomic Aging Clock) and validated across multi-modal epigenetic projections.

- **Safety Loss Constraint Engineering:** Designed neural loss functions enforcing >94% Troponin T and Connexin-43/GJA1 retention (0% arrhythmia risk) and oncogenic locks (OCT4 <= 0.35, c-MYC <= 0.30).
- **Robotics Automation Pipeline:** Constructed an automated script generator converting AI model outputs into 1-click executable CSV/Python scripts for Opentrons OT-2 robotics.
- **Accelerator & AI Ecosystem Leadership:** Leading deep-tech execution as part of Home Lab Accelerator Cohort 5.

PROMATLY & SAHARYN — London, United Kingdom

Founder & Chief AI Software Architect (2024 – Present)

- Architected Promatly.com, an enterprise AI automation platform executing Zapier integration apps, REST API bridges, and multi-agent AI workflow orchestration.
- Designed Saharyn.com, a real-time data analytics dashboard, cloud automation pipeline, and telemetry synchronization engine.

SWANSEA BAY UNIVERSITY HEALTH BOARD — Swansea, United Kingdom

- Cancer Clinical Trial Researcher (01/2024 – 05/2024)
 - Contributed to the analysis, design, and execution of oncology research protocols across trial phases.
 - Managed participant eligibility screening and led informed consent processes following strict GCP and NIHR standards.
 - Maintained complete Good Clinical Practice (GCP) compliance, ensuring total data reliability, patient safety, and regulatory audit readiness.
 - Conducted systematic medical literature reviews and collaborated within multidisciplinary clinical oncology teams.

PLAN INTERNATIONAL & WE LOVE MORRISTON — Sudan & United Kingdom

Healthcare & Community Development Volunteer (2017 – 2020)

- Provided COVID-19 health education, hygiene advocacy, and infectious disease prevention across local communities and refugee centres.

EDUCATION

BSc (Hons) Applied Medical Sciences (09/2020 – 12/2023)

Swansea University, Swansea, United Kingdom

Bachelor of Medicine and Surgery (MBBS / MD Studies) (11/2013 – 05/2019)

University of Al Fasher, Sudan

Neuroscience Training Course — Sudanese Society of Neurological Sciences (SSNS) 03/2017

Diploma of Clinical Pharmacy (01/2017 – 05/2017)

Nouran Academic Centre for Training and Capacity, Sudan

AWARDS, RECOGNITION & PATENTS

- **Silver Medal Winner — 50th International Exhibition of Inventions Geneva (Switzerland):** Invented the Real-Time Antigen-Antibody Detection (RTAAD) diagnostic device for rapid, high-sensitivity testing.
- **UK Patent Pending:** Filed in 2024 with the UK Intellectual Property Office (UK IPO).
- **UK Registered Design Rights:** Officially registered with the UK Intellectual Property Office.
- **Granted Patent Holder:** Granted for a novel medical testing device (2018).
- **2 Bronze Medals:** IPITEX International Intellectual Property & Innovation Expo (2024).
- **3 Silver Medals:** Awarded by the National Research Council of Thailand (NRCT) (2023).
- **British Council Winner:** Winner of the Sudan British Council Entrepreneurship Award (2019).
- **Stars of Science Finalist:** Top 30 Finalist in Stars of Science Qatar Foundation flagship competition (2019).
- **UNESCO Nominee:** Nominated for Innovation HUB, African Week in Paris (2019).

CERTIFICATIONS & ACCREDITATIONS

- Good Clinical Practice (GCP) Consolidation Training — NIHR, UK (03/2024)
- Research Human Tissue Act Certification — NIHR, UK (05/2024)
- Transporting Dangerous Goods & IATA Certification — Mayo Clinic Laboratories (05/2024)
- Valid Informed Consent in Research — Health and Care Research Wales (02/2024)
- Social Services and Well-being Act & Resuscitation — NHS Wales (01/2024)
- Safeguarding Adults and Children — NHS Wales (01/2024)
- Welsh Language Certification — NHS Wales (01/2024)